

How image reconstruction is like firing a cannon

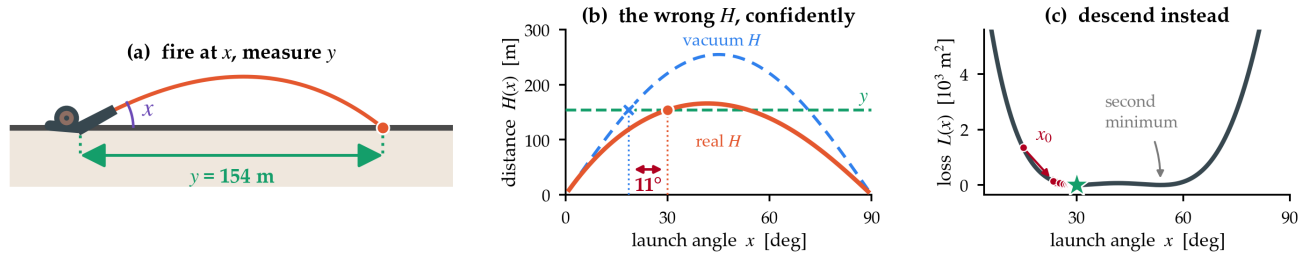
The same inverse problem, and the same two ways to solve it

Every reconstruction problem has the same shape: something you want to know, a physical process that turns it into something you can measure, and the job of running it backwards. A cannon is the smallest honest example.

The setup

You fire a cannonball at an angle x and it lands a distance y away. The laws of physics H turn the angle into the distance, $y = H(x)$, and that direction is easy: pick an angle, integrate the motion, read off where it lands. Now turn the question around. You want to land a shot a distance y away, so what angle do you aim at? Equivalently, you measured y , so what x produced it? That is the inverse problem, and an algorithm A that answers it is a reconstruction algorithm.

cannon	image reconstruction
x	the image you want to recover
y	the projection data you acquired
H	the system model: geometry, attenuation, scatter, blur
A	the reconstruction algorithm



(a) One measurement, one unknown. (b) Inverting the wrong forward model gives a confident, wrong answer. (c) No formula to invert, so descend instead.

Simplified physics: the inverse is a formula

In a vacuum the physics collapses to one expression, and you can invert it once, on paper:

$$y = H(x) = \frac{v^2 \sin(2x)}{g} \quad \Rightarrow \quad \hat{x} = A(y) = \frac{1}{2} \sin^{-1} \left(\frac{yg}{v^2} \right)$$

One line, instant, exact. Filtered back-projection is the same move: assume an idealised H (clean line integrals, no attenuation, no scatter), invert it analytically, then apply that inverse once.

Real physics: there is no formula

Now add air. Drag couples the two components of the motion through the speed, and the trajectory solves a pair of nonlinear differential equations. You can still *evaluate* H , integrating it to read off where the ball lands, but you cannot solve it for x . This is the ordinary situation in reconstruction: once H carries attenuation, scatter, collimator response and detector blur, H^{-1} has no closed form either. But you can always simulate forward and compare, scoring each candidate angle by how badly its simulation misses, then walking downhill:

$$\hat{x} = \operatorname{argmin}_x \frac{1}{2} (H(x) - y)^2 \quad x_{n+1} = x_n - \alpha (H(x_n) - y) \frac{\partial H}{\partial x}$$

The new guess is the old guess, minus a step size, times how wrong the simulation was, times how the measurement responds to what you are changing. Every iterative reconstruction algorithm is that loop: MLEM, OSEM, penalised likelihood, model-based. Only the cost function and the exact step change. Panel (c) shows the catch: the loss has two minima, because the same distance can be reached by firing low or by firing high, so which answer you reach depends on where you start.

The model is the whole game

The failure mode is not noise, it is the wrong H . Fire at 30° through air and the ball lands at 154 m; hand that number to the vacuum formula and it returns 19° , a precise and confident answer that is wrong by 11° , because the algorithm inverted a model that did not generate the data.

SPECT of high-energy emitters is exactly this. Image Ac-225 through its daughters, Bi-213's 440 keV line among them, and much of the signal reaches the crystal by penetrating or scattering in the collimator rather than passing cleanly through its holes, which a traditional H does not model. That H returns the 19° answer: a sharp, confident, quantitatively biased image that neither more counts nor a better optimiser will repair. The missing physics has to go into H .

*Your image has a million unknowns and the cannon has one, but the structure is identical.
Get H right and the rest is bookkeeping; get it wrong and no algorithm will save you.*